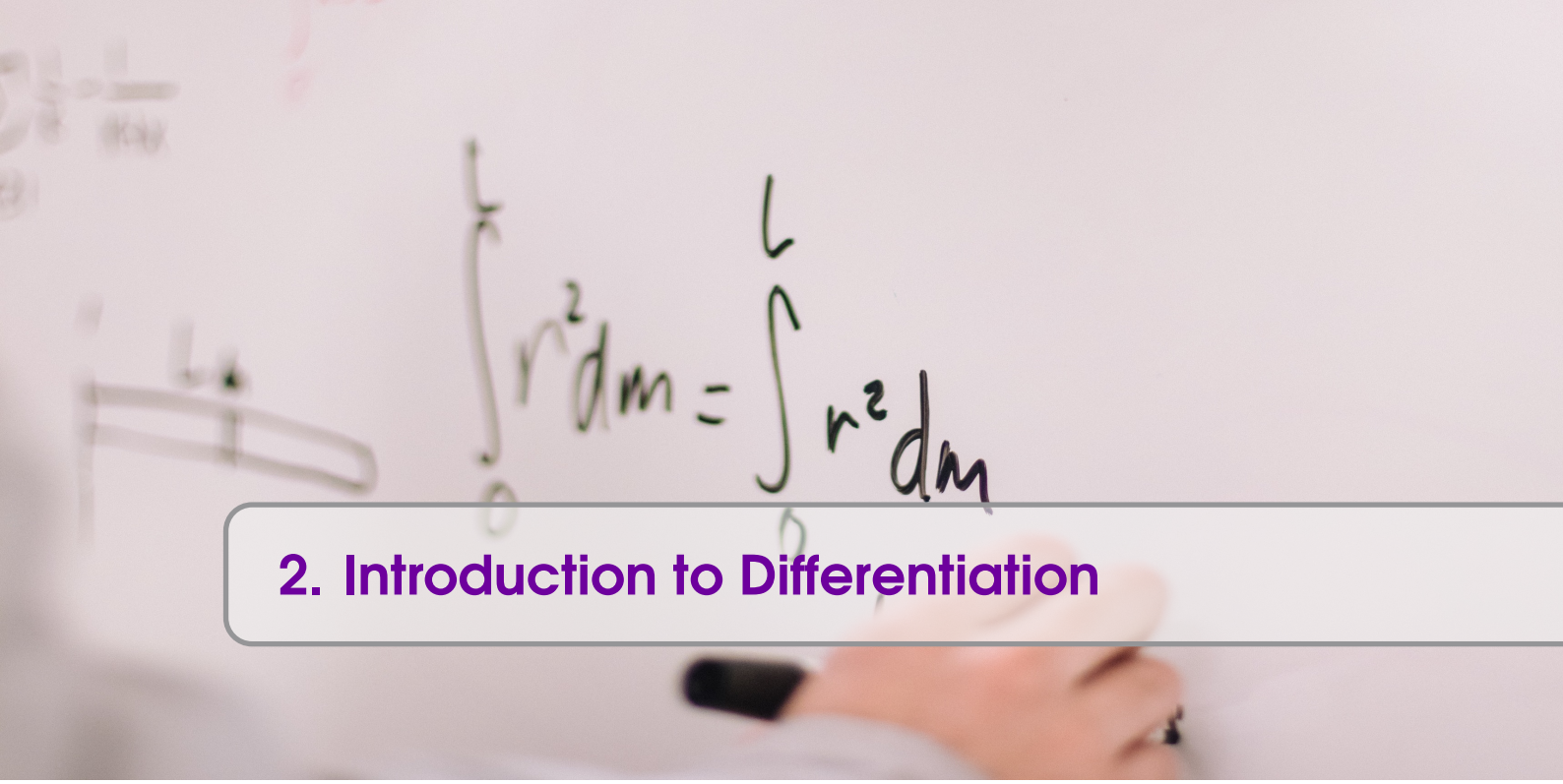


2. Introduction to Differentiation

2.1 Overview

Take a look at the graph of the function $f(x) = 2x^3 - 20x^2 + x + 4000$ in Figure 2.1. Clearly it can be seen that the gradient of this line is not constant with x. For example:

1. at $x = -5$ the gradient is quite steep and positive.
2. at $x = 0$ the gradient is almost zero.
3. at $x = 5$ the gradient is fairly shallow and negative.

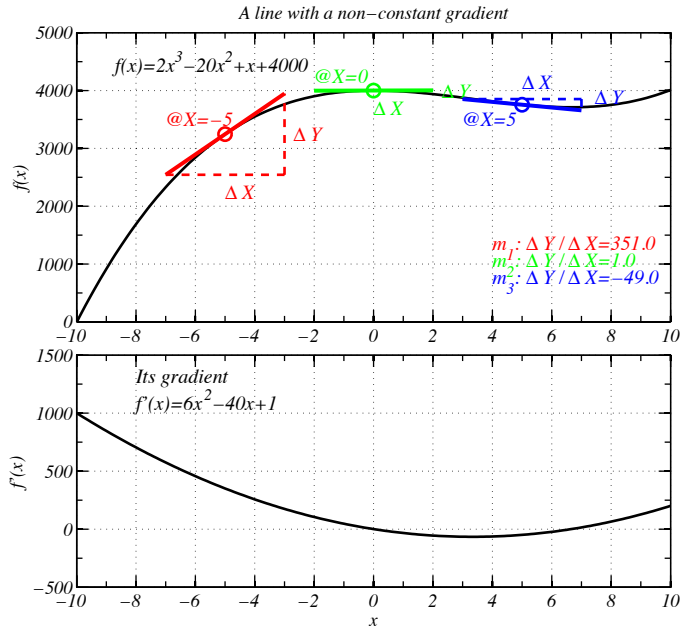


Figure 2.1: Top shows the function $f(x) = 2x^3 - 20x^2 + x * 4000$; bottom shows the gradient $\frac{dy}{dx}$, or $f'(x)$, which is positive until about $x = 0$, negative until about $x = 6$ and positive thereafter.

You could find the gradient at every point by drawing a tangent to the line at that point and determining how much y changes for a given change in x, like we did in Chapter 1. However, we can use **differentiation** to find a function that describes the gradient, which we will call $\frac{dy}{dx}$. In this case the function $6x^2 - 40x + 1$ describes the gradient of $f(x)$. This is shown in the bottom panel of Figure 2.1, labelled as $f'(x)$ which is often used in place of $\frac{dy}{dx}$. Once we have $\frac{dy}{dx}$, we can evaluate it for any value of x to find the gradient.

2.2 Defining differentiation

How can we arrive at the formula used to generate the line in the bottom figure? We use the rules of **differentiation**. These rules are central to the application of calculus. We do not cover the derivation of all rules here; rather we learn how and where to apply them. You should treat these rules as an engineer might treat a screwdriver or hammer. Please keep a copy of your crib sheet nearby throughout the course. You can use this to determine which rule to apply for a given problem that requires differentiation. For now, let us take a look at a straight line and quadratic.

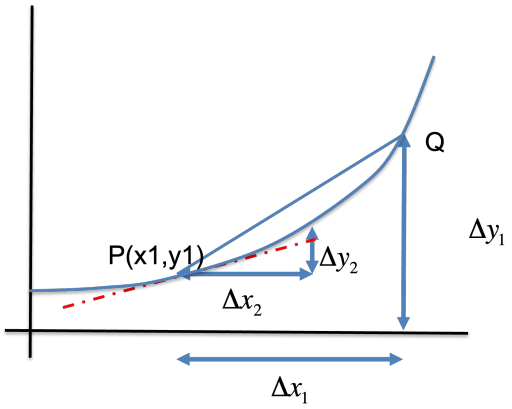


Figure 2.2: A hypothetical curve with a y value that increases non linearly as x increases. The gradient at points P and Q are different.

In Figure 2.2 we find a curved line and compare two positions on that line, namely P and Q. These position have coordinates, or values for x and y as (x, y) and $(x + \Delta x_1, y + \Delta y_1)$ respectively. In other words, position Q is found by movement along from P by an increment given by Δx_1 and Δy_1. We want to derive an expression for $\frac{dy}{dx}$, and we keep referring to approximating this using a finite difference approach (please refer to week 1). However, the choice of the increment used to arrive at this approximation is important. You can tell from our figure that we will arrive at a different gradient if we used a combination of increments Δx_1 and Δy_1 compared with Δx_2 and Δy_2. If this isn't clear, please ask in class. As our increment gets smaller and smaller, we get closer to the accurate value for $\frac{dy}{dx}$.

Indeed, we can write the following equation:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x}$$

which states that as our increment, Δx tends to 0, we arrive at our derivative value. This gives us a mechanism for deriving a formula for our gradient. For example, take the straight line displayed in Figure 2.3.

Derivative of a straight line

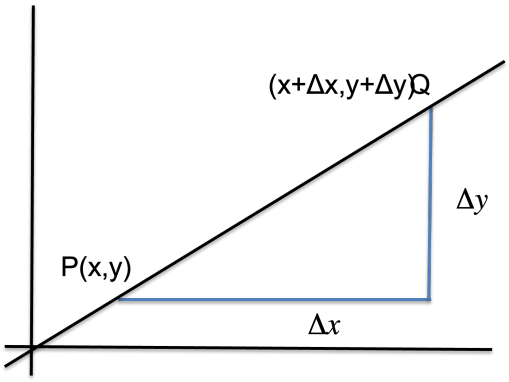


Figure 2.3: A hypothetical curve with a y value that increases linearly as x increases. The gradient at points P and Q are the same so we expect our gradient, thus derivative, to be constant.

The equation of a straight line can be expressed as the following:

$$y = ax$$

where the constant a is our gradient. We know our derivative $\frac{dy}{dx}$ should therefore be equal to a. But can we show this through a derivation? Using the equation of the straight line we can work through the following steps to arrive at an expression for $\frac{\Delta y}{\Delta x}$:

$$\begin{aligned} y + \Delta y &= a\,(x + \Delta x) \\ &= ax + a\Delta x \end{aligned}$$

If we then subtract $y = ax$ from both sides we get :

$$\Delta y = a\Delta x$$

Rearranging this gives us:

$$\frac{\Delta y}{\Delta x} = a$$

If we let $\Delta x \mapsto 0$ then we see that

$$\frac{\Delta y}{\Delta x} \to \frac{dy}{dx} = a$$

So perhaps if we can keep deriving expressions for $\frac{\Delta y}{\Delta x}$ we can see if that expression simplifies as Δx tends to 0 and identify a pattern which we could wrap up in a general rule.

Derivative of a quadratic line

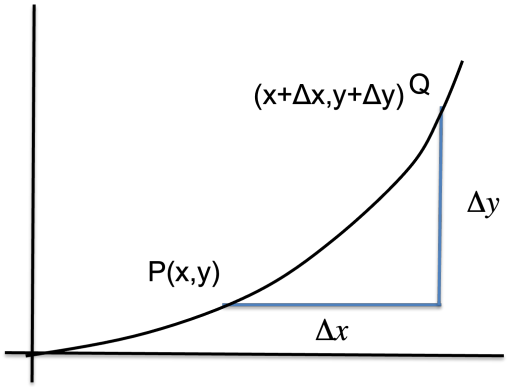


Figure 2.4: A hypothetical curve with a y value that increases quadratically as x increases. The gradient at points P and Q are different.

The equation of a quadratic can be expressed as the following:

$$y = ax^2$$

In Figure 2.4 we again position two points on our curve. Let us derive an expression for $\frac{\Delta y}{\Delta x}$ in moving between these points by substituting the x and y coordinates and isolating expressions for Δx and Δy:

$$\begin{aligned} y + \Delta y &= a(x + \Delta x)^2 \\ &= ax^2 + 2ax\Delta x + \Delta x^2 \end{aligned}$$

Lets subtract $y = ax^2$ from both sides of the equation:

$$\Delta y = 2ax\Delta x + \Delta x^2$$

Now let's divide the equation by Δx:

$$\frac{\Delta y}{\Delta x} = 2ax + \Delta x$$

If we let $\Delta x \mapsto 0$ then we see that

$$\frac{\Delta y}{\Delta x} \to \frac{dy}{dx} = 2ax$$

If we repeat this procedure for higher order polynomials, we calculate that the gradient function for some simple functions of x by **multiplying the original function of x by its power** and **reducing the power that x is raise to by 1**:

$$\begin{aligned} y = x^n &\implies \frac{dy}{dx} = nx^{n-1} \\ y = ax^n &\implies \frac{dy}{dx} = anx^{n-1} \end{aligned}$$

2.3 Common differentials

What if we didn't have a polynomial in x? There are many other types of functions, so we will also need expressions for other common types of functions. Table 2.1 we list all of the common differentials you will use in this course. Learning how to apply them is a key aspect of the problems we will encounter.

Table 2.1: List of common differentials

Function $y = f(x)$	**Derivative** $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

2.4 Differentiating sums of functions

So far we have seen how to differentiate simple functions (see Table 2.1. What happens when we have a sum of functions such as:

$$y = 2x^2 + 3x$$

It turns out that differentiation is invariant to sums of functions. This means that when you differentiate the sum of two or more functions:

$$y = f(x) + g(x)$$

the **derivative of the sum** is equal to the **sum of the derivatives** of the individual functions.

$$\frac{dy}{dx} = \frac{df}{dx} + \frac{dg}{dx}$$

Therefore, using the list of common derivatives, we see that the derivative of $y = 2x^2 + 3x$ is:

$$\frac{dy}{dx} = 2x + 3$$

We'll see what happens when we get more complicated combinations of functions (i.e. functions of functions, products, fractions) in the next Chapter!